Handoff 01 — PCB final review and order

For: Claude Code in VS Code + KiCad 9 GUI, with Matt. ~45 min. **Matt places the order.**

The boards were generated, routed and checked in the cloud (KiCad 9.0.9, kicad-cli): ERC 0, DRC 0 errors, 0 unconnected, schematic parity clean. That run was after verification round 1; see `docs/verification/round1.md` for what changed (Q4 LED-board switch, F1 PTC, R3 3.9 k, R5 10 k, R11 VREF, J2 moved with a lead slot, LM75B thermostat wiring), then round 2 (`docs/verification/round2.md`: battery-lead slot 6.4 × 5.0 mm, Q4 soft turn-on R13/C8, 1.1 A PTC, VLED test pad TP7, RESET pull-up 2.4 k, W/C silk next to the white LEDs). The fab packages are ready in `fab/main-board/v0.1/` and `fab/led-board/v0.1/`. The DRC warnings that remain are silk-only, and the reports list them: the main board's SW1 outline is clipped at the board edge, because the knob overhangs the edge by design. This handoff is the human look-over the pcb repo's process requires (steps 3, 4, 6, 7 of its fabrication process), then ordering.

1. Open and inspect (KiCad GUI)

1. `git pull`; open `hardware/main-board/main-board.kicad_pro` and `hardware/led-board/led-board.kicad_pro`.
2. **Do not edit and save the generated files** unless you intend to stop regenerating that board (see CLAUDE.md).
3. Run ERC and DRC in the GUI; expect the same results as `hardware/*/reports/*.json`.
4. Open the 3D viewer (Alt+3) and check:
    - Every part is oriented plausibly.
    - Main board: the USB-C mouth is at the bottom edge, and the slide-switch knob sits past the right edge. The JST-SH opening faces the top edge. The JST-PH (battery) opening faces into the board, and the lead slot is beside it.
    - LED board: 1.0 mm thick, with the LEDs on top and the TLC59116, LM75B and G/V/D/C pads underneath.
    - The cloud build had no KiCad 3D models for J1, J2, F1 and the 2835 LEDs, so they are missing from the committed STEP files (round2 R2-D17). Re-export the STEP on your machine (`kicad-cli pcb export step --subst-models`) before using it for fit checks. J2 is the tallest part, at 5.5 mm.
5. **1:1 paper print** of both boards (File → Print, 100 %, check with a ruler). Sit real parts on it if they have arrived: tact switch, USB-C, PH header, 2835 LED. Check the LED board against the printed moon-base cup pocket.

2. Gerber check

1. Open `fab/<board>/v0.1/<board>-v0.1-gerbers.zip` in KiCad's Gerber viewer: one layer at a time.
2. Upload the same zip to the fab's online viewer and compare.

3. Order (Matt)

Board specs (also in each `FAB_NOTES.md`):

	Main board	**LED board**
Size	35 × 62 mm	Ø 26 mm round

	Main board	LED board
Layers / thickness	2 / 1.6 mm FR4	2 / **1.0 mm** FR4
Copper	1 oz	1 oz
Finish	HASL lead-free (ENIG optional)	**ENIG** (flat pads for LEDs/TSSOP)
Mask / silk	green / white	**white / black** (white mask reflects light into the moon)
Min track/space, via	0.2/0.2 mm, 0.6/0.3 mm	0.2/0.15 mm, 0.5/0.25 mm
Qty	5	5
Stencil	yes (frameless, top only)	yes (frameless, **top and bottom**)

- **Route chosen: bare boards + stencils, hand assembly with Farnell parts** (see `bom/assembly_routes.md` for the price comparison of all three routes). If Matt prefers assembly instead: upload `bom_pcbway.csv` + `cpl_pcbway.csv` (PCBWay) or `bom_jlcpcb.csv` + `cpl_jlcpcb.csv` (JLCPCB; fill the LCSC column from their search), and check every part's rotation in the fab's placement preview (KiCad zero-orientation ≠ reel orientation). Small boards may need panelising for assembly at PCBWay (main 35 × 62 mm, LED Ø26 mm).
- After ordering: commit with the order reference in the message, tag `main-board-v0.1` / `led-board-v0.1`.

Done when

Both boards ordered (or a decision recorded not to), order references committed, tags pushed.